

Performance Evaluation of Wave Energy Converters: A Two-Phase Data-Driven Methodology

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I. INTRODUCTION

The commercial viability of offshore renewable energy, particularly Wave Energy Converters (WECs), is heavily dependent on minimizing Operation and Maintenance (O&M) costs to reduce the Levelized Cost of Energy (LCOE). A primary challenge in WEC fleet management is distinguishing between normal performance fluctuations caused by atypical metocean conditions (environmental false positives) and genuine mechanical degradation (e.g., Power Take-Off (PTO) failures) [2]. Aligning with the TEF RES Service 2 framework, this work presents a robust, data-driven methodology to continuously monitor and evaluate the operational efficiency of a WEC farm.

The studied architecture monitors twelve C5 WECs, each with a 350 kW nominal capacity, deployed within a 6 km radius. To accurately isolate mechanical faults from environmental noise, we propose a methodology structured in three sequential phases:

- **Phase 1 (Absolute Analysis):** A Machine Learning Digital Twin using Extreme Gradient Boosting (XGBoost) to establish an absolute performance baseline [3].
- **Phase 2 (Relative Benchmarking):** A cross-asset evaluation using Stochastic Frontier Analysis (SFA) to decompose energy deficits into symmetric environmental noise and one-sided technical inefficiency [5].
- **Phase 3 (Decision Engine):** A deterministic integration matrix that merges absolute residuals and relative efficiency to actively suppress false alarms and map operational states.

II. SCOPE AND CORE ASSUMPTIONS

The methodology relies on strictly defined physical and statistical boundaries to ensure diagnostic accuracy across the pipeline.

A. Sets and Indices

- **Assets:** Identical WEC units $i \in I = \{1, 2, \dots, 12\}$.
- **Time:** Evaluated over discrete 30-minute high-frequency intervals $t \in T$.

B. Deterministic Parameters

- **Power Limit:** A hard physical constraint on the maximum generation output per unit, $P_{max} = 350$ kW.
- **Spatial Homogeneity:** The maximum farm radius of 6 km guarantees that all assets are exposed to highly correlated, macroscopic metocean conditions simultaneously.

C. Stochastic Drivers

- **Wave Power Flux:** The dominant physical predictor for energy conversion, defined as $WPF_{i,t} \approx 0.49 \times H_s^2 \times T_e$, which compounds Significant Wave Height (H_s) and Energy Period (T_e).

D. System Variables

The pipeline continuously computes three critical state variables per asset:

- **Predicted Power Output ($\hat{Y}_{i,t}$):** The absolute generation baseline expected by the XGBoost Digital Twin (Phase 1).
- **Technical Efficiency Score ($TE_{i,t} \in [0, 1]$):** The relative performance metric computed via the Battese-Coelli (1988) SFA estimator (Phase 2).
- **Operational State ($S_{i,t} \in \{0, 1, 2, 3\}$):** The final diagnostic classification assigned by the deterministic logic matrix (Phase 3).

E. Boundary Constraints

- **Safety Limits:** To prevent algorithmic extrapolation and non-physical predictions during extreme storm events, all power estimates and measurements are rigidly bounded: $0 \leq \hat{Y}_{i,t} \leq P_{max}$.

III. 3-PHASE METHODOLOGY & RESULTS

A. Phase 1: Absolute Analysis (Digital Twin)

Phase 1 establishes an empirical baseline of healthy operations using an Extreme Gradient Boosting (XGBoost) regressor to predict expected power output based on real-time sea states [4].

Methodological Core:

- **Data Preprocessing:** Handled missing sensor data (imputing 777 values via a median strategy) and trained the model on 15 distinct physical and temporal features.
- **Temporal Split:** The architecture utilized a strict 80/20 chronological split (69,580 training samples; 17,396 test samples) to prevent temporal data leakage.

- **Dynamic Confidence Band:** Predictions include a confidence band driven by the baseline model error.
- **Anomaly Threshold:** An absolute anomaly is flagged when the generation deficit strictly exceeds the expected baseline error margin: $Residual < -1.5 \times RMSE_{baseline}$.

Results (Phase 1):

- **Predictive Performance:** The regressor achieved a global out-of-sample R^2 of 0.4891, with a Mean Absolute Error (MAE) of 22.10 kW and an RMSE of 38.31 kW. Crucially, isolating Epoch 3 reveals an R^2 of 0.8844 for the healthy fleet (demonstrating the model retains high accuracy during nominal operations), while the degraded fleet R^2 collapses to -6.5206, mathematically confirming the severe anomalies.
- **Feature Dominance:** *Wave Power Flux* emerged as the overwhelmingly dominant predictor ($\sim 83.5\%$ importance), followed by Significant Wave Height ($\sim 14.3\%$).
- **Precise Degradation Flagging:** Using the strict baseline threshold (-5.86 kW), the model successfully isolated the Epoch 3 anomalies. Healthy WECs (Buoys 1-8) showed negligible anomaly rates ($< 0.5\%$), whereas the degraded WECs (Buoys 9-12) triggered alarms in nearly all affected timestamps, with Boia 12 peaking at a 99.02% anomaly rate.

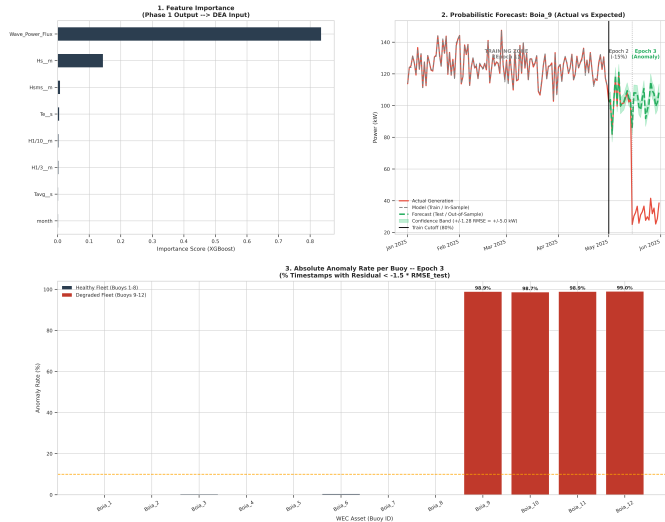


Fig. 1. Phase 1 Absolute Analysis: Feature importance (left) and Probabilistic Forecast detailing the dynamic confidence band against actual production (right).

B. Phase 2: Relative Benchmarking (SFA)

Phase 2 isolates true mechanical faults by comparing assets against each other using a Cobb-Douglas Stochastic Frontier Analysis (SFA). Unlike deterministic models, SFA decomposes the composite residual (ϵ) into two distinct statistical distributions.

Statistical Decomposition:

- **Symmetric Noise ($v_i \sim N(0, \sigma_v^2)$):** Represents natural sea-state variation, sensor noise, and measurement error.
- **Technical Inefficiency ($u_i \sim |N(0, \sigma_u^2)|$):** A strictly non-negative term capturing unidirectional energy loss (e.g., mechanical PTO failure).

Diagnostic Logic (The Merge):

- **False Positive (Sea State):** XGBoost flags an anomaly, but SFA TE is uniform across the farm. Machine is healthy; weather is atypical.
- **True Anomaly (Fault):** XGBoost flags an anomaly AND the SFA TE drops severely relative to the frontier.

Results (Phase 2):

- **Golden Frontier Calibration (Epoch 1):** Maximum Likelihood Estimation (MLE) successfully converged on healthy data ($NLL = -51116.18$), establishing a robust production surface baseline with a mean Technical Efficiency (TE) of ~ 0.88 across all assets.
- **Environmental Resilience (Epoch 2):** A 15% global performance drop was correctly absorbed by the model. TE shifted uniformly across the fleet to ~ 0.71 – 0.74 without isolating single buoys, proving resilience to environmental false positives.
- **Fault Isolation (Epoch 3):** Upon simulating a severe PTO failure, SFA Technical Efficiency for the degraded assets (Buoys 9 to 12) plummeted to ~ 0.26 – 0.28 (e.g., Boia 12 operating at exactly 28.0% of the frontier capability). In contrast, the healthy fleet remained stable at ~ 0.87 – 0.89 .
- **Generation Deficit Mapping:** The methodology dynamically calculated the physical energy loss attributable to mechanical inefficiency (u). In Epoch 3, degraded WECs exhibited severe deficits (117.67 kW to 133.71 kW per observation), sharply isolating them from the healthy fleet (< 14 kW deficit).

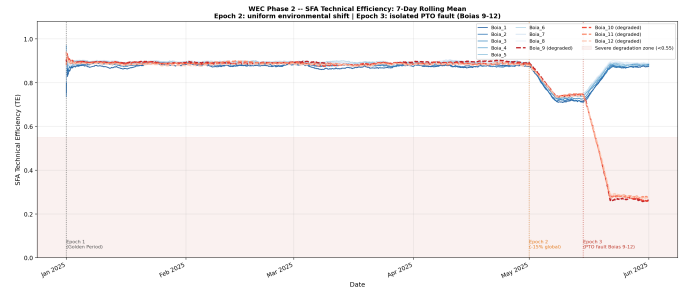


Fig. 2. Phase 2 7-Day Rolling Technical Efficiency: Demonstrates the uniform environmental shift in Epoch 2 and the isolated severe mechanical degradation of Buoys 9-12 in Epoch 3.

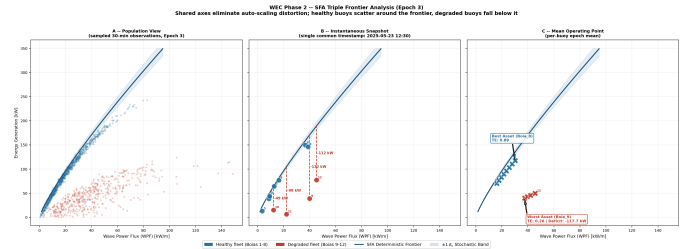


Fig. 3. Phase 2 Relative Analysis: SFA deterministic frontier and stochastic band. Degraded WECs visually detach from the healthy population cloud.

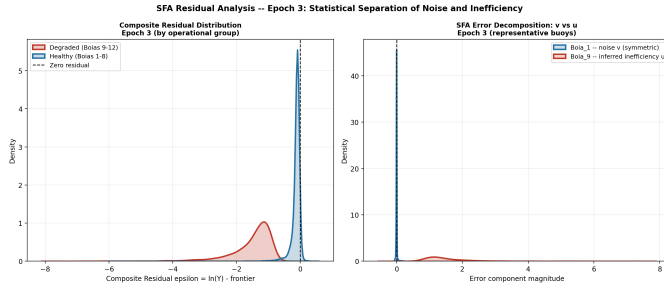


Fig. 4. SFA Residual Decomposition: Statistical separation of symmetric noise (v) from one-sided technical inefficiency (u), proving exact fault isolation.

C. Phase 3: The Merge (Decision Engine)

Phase 3 acts as the ultimate diagnostic engine, intersecting the absolute predictive capabilities of the Digital Twin (Phase 1) with the relative Production Possibility Frontier (Phase 2). This intersection addresses the primary operational challenge in offshore renewable energy: the eradication of false positives caused by atypical metocean conditions.

Methodological Core:

- **Spatio-Temporal Join:** The algorithmic pipeline merges the XGBoost predictions and SFA technical efficiency scores using high-frequency timestamps (t) and unique WEC identifiers (i) as primary keys.
- **Instantaneous Scoring Matrix:** For each observation, a deterministic logic gate evaluates two distinct conditions:
 - **Condition A (Absolute):** $Residual_{i,t} < -1.0 \times RMSE_{baseline}$
 - **Condition B (Relative):** $TE_{i,t} < 0.60$ (SFA Efficiency threshold)
- **Direct Fault Aggregation:** By removing complex temporal smoothing layers, the engine strictly aggregates the total hours each WEC spends in a state of simultaneous absolute and relative failure, providing a direct metric for O&M dispatch prioritization.

Diagnostic State Matrix: Depending on the instantaneous evaluation of Conditions A and B, the system classifies the asset into one of four operational states:

- **State 0 (Nominal):** Both conditions FALSE. The WEC is healthy and operating exactly as expected.
- **State 1 (Environmental False Positive):** Condition A TRUE, Condition B FALSE. The asset underperforms its historical baseline but maintains its relative position on the farm's frontier. The algorithm overrides the absolute alarm, attributing the drop to external phenomena.
- **State 2 (Latent Degradation):** Condition A FALSE, Condition B TRUE. The asset begins to deviate from its peers on the SFA frontier, but the absolute generation deficit has not yet breached the RMSE threshold.
- **State 3 (Critical Fault):** Both conditions TRUE. The asset falls below both its historical baseline and the instantaneous frontier, confirming a mechanical anomaly.

Results and Validation (Phase 3): The integration pipeline was executed on the complete dataset containing 86,122 synchronized observations, yielding the following results:

- **Global Fleet Distribution:** The fleet operated in nominal conditions (*State 0*) during 80.61% of the analyzed time-frame (69,423 observations).
- **False Positive Elimination:** The system successfully identified 12,162 instances of environmental false positives (*State 1*, 14.12%). By blocking these absolute alarms, the decision engine prevented unnecessary offshore vessel deployments during severe sea states, directly optimizing the farm's operational costs.
- **Latent Fault Tracking:** The SFA frontier mapped 980 instances of latent degradation (*State 2*, 1.14%), identifying the early stages of asset deviation before they could breach absolute historical error margins.
- **Targeted O&M Dispatch (Epoch 3):** By aggregating the hours spent strictly in Critical Fault (*State 3*), the system distinctly separated the healthy fleet from the degraded fleet. Healthy assets (Buoys 1-8) registered negligible critical downtime (0.5 to 4.5 hours due to transient signal noise), whereas the degraded cluster (Buoys 9-12) accumulated massive critical fault hours, prompting immediate maintenance dispatch:
 - **Boia 11:** 339.5 hours of Critical Fault.
 - **Boia 12:** 334.0 hours of Critical Fault.
 - **Boia 10:** 332.0 hours of Critical Fault.
 - **Boia 9:** 321.0 hours of Critical Fault.

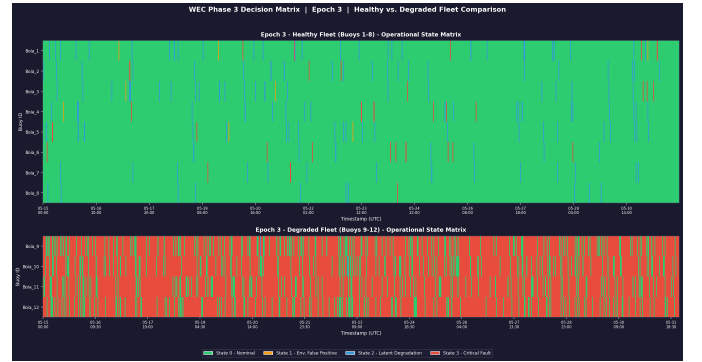


Fig. 5. Phase 3 Decision Matrix Heatmap (Epoch 3): Comparison between the healthy fleet (Buoys 1-8, predominantly nominal State 0) and the degraded fleet (Buoys 9-12, dominated by critical State 3 faults).

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